Haojie Hu (胡豪杰)
Postdoctoral Researcher · JSPS Fellow
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Experience & Education

- **Postdoctoral Researcher, University of Tsukuba (JSPS Fellow)** — *Aug. 2024–Present*
  Astrophysics — Host: Prof. Ken Ohsuga
- **Ph.D. in Astrophysics, Peking University** — *Sep. 2018 - Jun. 2024*
  Major: Astrophysics — Advisor: Prof. Kohei Inayoshi
- **Teaching Assistant: Peking University** — *Sep. 2020–Jun. 2023*
  Stellar Formation and Evolution; Fluid Dynamics in Astrophysics
- **Summer School: Peking University** — *Jul. 2022*
  "Chinese Space Station Telescope" Summer School
- **B.S.: East China University of Science and Technology** — *Sep. 2014 - Jun. 2018*
  Major: Applied Physics — Advisor: Prof. Wei Liao

Research Interests

Black-hole accretion; super-Eddington accretion; outflows, winds, and jets; supermassive black holes; AGN feedback; SMBH–galaxy co-evolution; high-redshift quasars; JWST and XRISM.

Publications

6 refereed papers (5 first-author and 1 collaborative); 1 manuscript under review

1. **Hu H.**, Yanagisawa H., Nishigaki M., Kiyota T., Ishiyama T., Ohsuga K., Back to Normal Again: Possible Destinies of JWST overmassive SMBHs and "Little Red Dots" in the View of Shin-Uchuu Simulation. arXiv:2602.14496 (2026). Under review.

2. **Hu H.**, Asahina Y., Yoshioka S., Takahashi H. R., Ohsuga K., Clumpy outflows from super-Eddington accreting black holes. I. Radiation hydrodynamics simulations and observational implications. PASJ, 78(2), 524–535 (2026).

3. **HuH.**, Inayoshi K., Haiman Z., Ho L. C., Ohsuga K., The Convergence of Heavy and Light Seeds to Overmassive Black Holes at Cosmic Dawn. 2025, The ApJ Letter, 983, L37 (2025).

4. **Hu, H.**, Inayoshi, K., Haiman, Z., Quataert, E. & Kuiper, R. Long-term Evolution of Supercritical Black Hole Accretion with Outflows: A Subgrid Feedback Model for Cosmological Simulations. The ApJ. 934, 132 (2022).

5. **Hu, H.**, Inayoshi, K., Haiman, Z., Quataert, E. Li, W., & Kuiper, R. Supercritical Growth Pathway to Overmassive Black Holes at Cosmic Dawn: Coevolution with Massive Quasar Hosts. The ApJ. 935, 140 (2022).

6. **Hu, H.**, Qiu, Y., Gendron-Marsolais, M., Bogdanović, T., Hlavacek-Larrondo, J., Ho, L. C., Inayoshi, K., & McNamara, B. R. Signature of Supersonic Turbulence in Galaxy Clusters Revealed by AGN-driven Hα Filaments. The ApJ Letter, 929, L30 (2022).

7. Qiu, Y., **Hu, H.**, Inayoshi, K., Ho, L. C., Bogdanović, T., & McNamara, B. R. Dynamics and Morphology of Cold Gas in Fast, Radiatively Cooling Outflows: Constraining AGN Energetics with Horseshoes. The ApJ Letter, 917, L7 (2021).

Academic Service

Referee: *Publications of the Astronomical Society of Japan*; *The Astrophysical Journal Letters*.
Membership: Astronomical Society of China.

Technical Skills

Languages: English & Chinese
Programming: C/C++, Python, Fortran
Scientific Computing: Linux, LaTeX, VisIt, Cloudy, MATLAB, `galight` image decomposition
Courses: Accretion power in astrophysics, The first galaxies in the Universe, Cosmology, Galaxy structure and evolution, Galactic dynamics, Physics of the Interstellar And Intergalactic medium.
Simulation Codes: PLUTO, Enzo, INAZUMA/UWABAMI, Athena++; radiative transfer with M1 closure, variable Eddington tensor, ray tracing, and flux-limited diffusion

Research Projects

•Super-Eddington Accretion

Radiation-hydrodynamic simulations with PLUTO; flux-limited diffusion, ray tracing, and nonequilibrium thermochemistry

- Modeled the long-term spatial structure and temporal evolution of super-Eddington flows, quantifying how accretion-driven mass loss regulates the inflow reaching the black hole.
- Developed a physically motivated subgrid feedback prescription for cosmological and galaxy-scale simulations (Paper 4).
- Ongoing work resolves outflow-regulated accretion from Bondi to bulge scales and its role in rapid black-hole growth during bulge assembly; one manuscript is in preparation.

•Outflows from Accretion and AGN Feedback

3D Enzo/AMR simulations and general-relativistic radiation-hydrodynamic simulations with INAZUMA

- Connected accretion-scale winds to galaxy- and cluster-scale feedback by following the transport of energy and momentum across physical scales.
- Reproduced the morphology and velocity structure of radiatively cooling Hα filaments in the Perseus cluster, constraining the energetics and history of AGN activity (Papers 6–7).
- Investigated the formation, acceleration, and observational signatures of clumpy radiation-driven winds near super-Eddington black holes, motivated by XRISM spectroscopy (Paper 2; one manuscript in preparation).

•Semi-Analytic Models of SMBH Growth and Galaxy Co-evolution

Simulation-calibrated semi-analytic modeling based on dark-matter halo merger trees

- Explored super-Eddington growth pathways for light and heavy seeds and their convergence toward overmassive SMBHs at cosmic dawn.
- Followed the coupled evolution of black holes and host galaxies to $z = 0$, testing how feedback regulates the black-hole-to-galaxy mass ratio.
- Interpreted JWST overmassive SMBHs and "Little Red Dots" and developed testable demographic predictions (Papers 1, 3, and 5).

Seminar & Contributed Talks

•Contributed Talks

- BHJET2026 Workshop, 11–13 March 2026, Komazawa University, Tokyo, Japan
- XRISM International Conference 2025, 20–25 October 2025, Kyoto, Japan
- Galaxy-IGM Workshop 2025, Waseda University, Japan (Hotel Sun Valley Nasu), 22-25, July 2025 (Best Group Work on Galaxy Theory)
- BH accretion workshop 2025, Nagoya City University (5-7, May 2025)
- Probing the Genesis of Supermassive Black Holes: Emerging Perspectives from JWST and Expectation toward New Wide-Field Survey Observations Kavli IPMU (18-21, Nov. 2024)
- First Star VI in NYC (20-23, May 2024)
- COSPAR in Busan (13-21, Jul. 2024)
- Galaxy formation in Hangzhou: Observation and Physics of AGN, China, Oct. 2023
- East Asian AGN workshop at Kagoshima, Japan, Sep. 2023
- Astrophysical BHs: A Rapidly Moving Field at Hong Kong, China, Jun. 2023
- Galaxy-IGM workshop 2022 at Japan, Aug. 2022
- 2021 The first KIAA forum on gas in galaxies for early career scientist, Nov. 2021
- East-Asia AGN workshop in Chongqing University, China, Oct. 2021

•Seminar Talks

- Seminar talk @ Center for Interdisciplinary Theoretical and Mathematical Sciences/RIKEN, 10 April 2026
- Seminar talk @ Zhejiang University, May 2023
- Seminar talk @ Shanghai Astronomical Observatory, CAS, Shanghai, China, May 2023
- Seminar talk @ Tsung-Dao Lee Institute, Shanghai Jiao Tong University, Shanghai, May 2023
- Seminar talk @ Tsinghua University, Beijing, China, Mar. 2023
- Seminar talk @ University of Duisburg-Essen, Oct. 2022
- Seminar talk @ Kavli IPMU, Jun. 2022